MATHS TEST

Class – IX (UP Board Pattern)

Chapters: Real Numbers & Algebraic Identities

Time Allowed: 1 Hour 30 Minutes

Maximum Marks: 35

General Instructions:

1. All questions are compulsory.
2. Use of calculator is not permitted.
3. Show necessary steps for full marks.

Section – A ($1 \times 9 = 9$ Marks)

1. Which of the following numbers is irrational?

(a) 0.16 (b) $\sqrt{25}$

(c) $\sqrt{11}$ (d) 7/9

2. The decimal expansion of 13/50 is:

(a) Terminating (b) Non-terminating recurring

(c) Non-terminating non-recurring (d) Irrational

3. The value of $(a + b)^2 – (a – b)^2$ is:

(a) 2ab (b) 4ab

(c) $a^2 – b^2$ (d) 0

4. Rationalising factor of $(3 + \sqrt{5})$ is:

(a) $3 + \sqrt{5}$ (b) $3 – \sqrt{5}$

(c) $\sqrt{5} – 3$ (d) $1/(3 + \sqrt{5})$

True / False

5. Every integer is a rational number. (True / False)

6. $\sqrt{3}$ is a rational number. (True / False)

7. $(a – b)^2 = a^2 – b^2$ (True / False)

Fill in the blanks

8. Decimal expansion of 1/16 is _______.

9. $(a + b)(a – b)$ = _______.

Section – B ($2 \times 3 = 6$ Marks)

10. Convert 0.4444… into p/q form.

11. Evaluate 103×97 using suitable identity.

12. If $x + y = 9$ and $xy = 20$, find $x^2 + y^2$.

Section – C ($3 \times 4 = 12$ Marks)

13. Convert 0.363636… into p/q form.

14. Rationalise $5/(\sqrt{7} + 2)$.

15. If $x + 1/x = 5$, find $x^2 + 1/x^2$.

16. Expand $(a + b + c)^2$.

Section – D (4 × 2 = 8 Marks)

17. Prove that $\sqrt{5}$ is irrational.

18. If $x + y + z = 0$, prove that $x^3 + y^3 + z^3 = 3xyz$.